



Clinical characteristics of **antidepressant treatment escalation** in major depressive disorder in All of Us

Sina Mahdiani

HGEN603

Spring 2026

+300 M

+29 M

1 out of 3

20 times

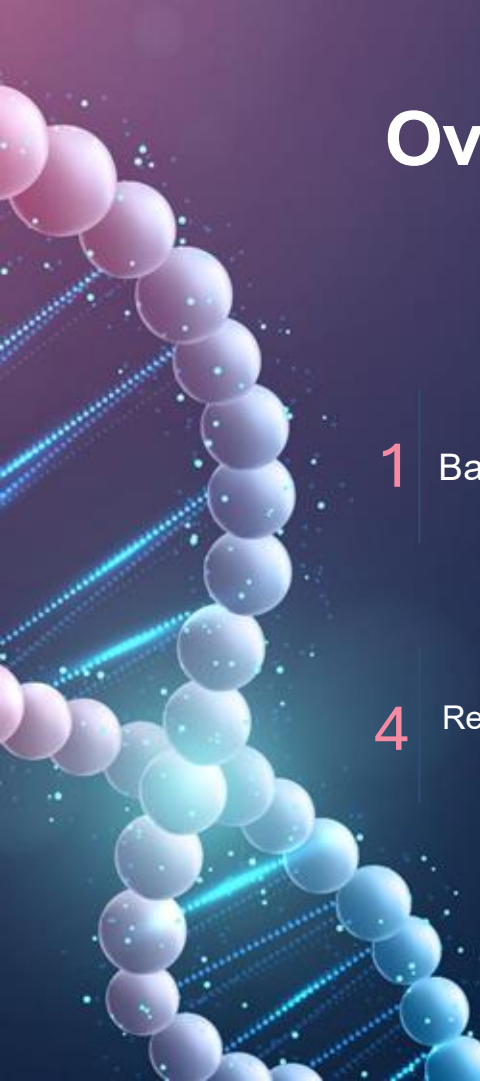
Twice

2 out of 3

\$92.7B

\$43.8B





Overview

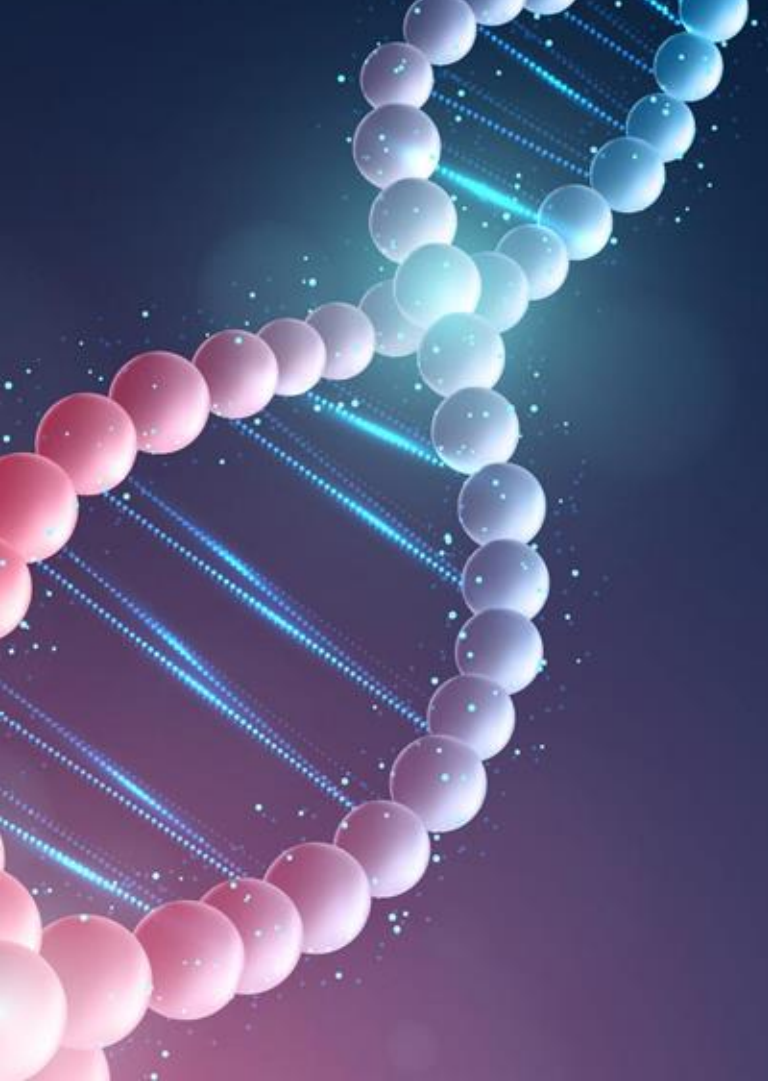
1 Background

2 Hypothesis and research question

3 Methods

4 Results

5 Discussion



INTRODUCTION

TRD: The Stubborn Depression

Different sources, different literatures provide different definitions ...

- Generally, it refers to major depressive episodes that do not respond satisfactorily to **at least two trials** of antidepressant monotherapy (AD)
- Any single failure of **an adequate trial (6–8 weeks)** of an AD
- if patients experience a less than **50% reduction** in depressive symptoms after **4 weeks**
- patient who does not respond to **pharmacotherapy** for at least **6 to 8 weeks** or to a **psychotherapy** trial for a minimum of **10 to 18 weeks**

TRD is different from severe MDD





How to check the severity?

PHQ-9 : score ≥ 20

DSM-5-T: have eight or nine

HAM-D : score ≥ 19



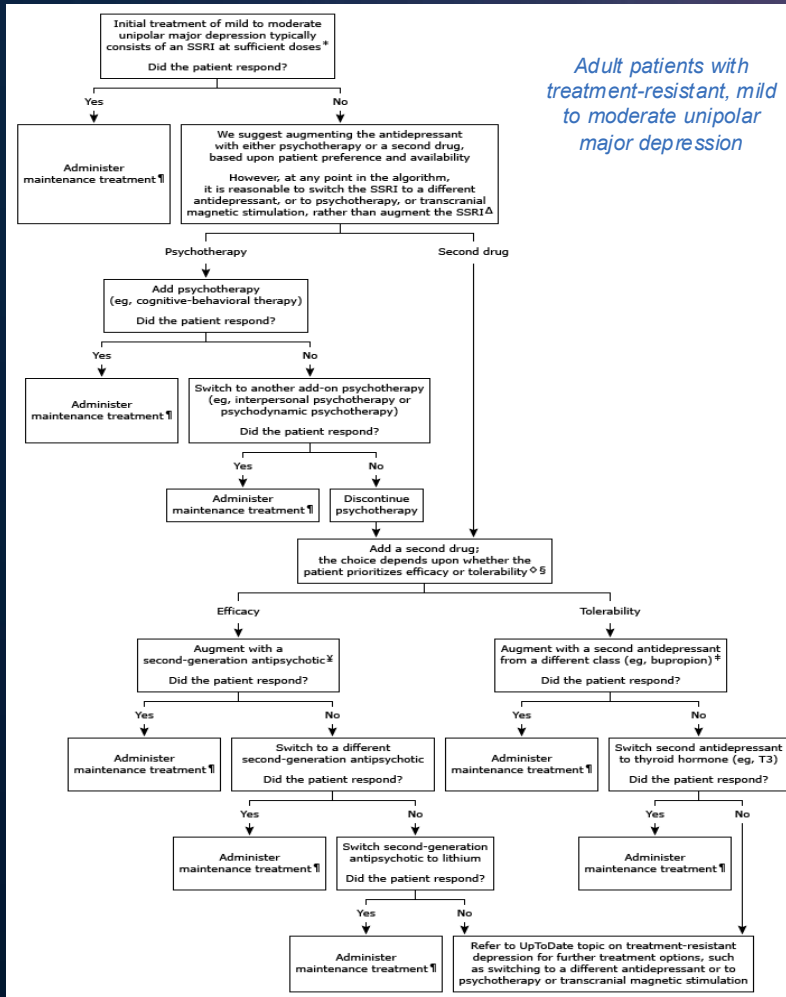
Medical options for these patients

ketamine/Esketamine

- Substance use disorder
- Dissociation and psychotomimetic effects

Electroconvulsive therapy

- memory impairment
- Musculoskeletal effects



What are the **challenges** in the treatment of TRD?

1. Lack of a Standard Definition!
2. Complex and Multifactorial Pathophysiology
3. Limited Clinical and Preclinical Biomarkers
4. Treatment Inefficacy and Side Effects
5. Comorbidities and Social Factors



Systematic review and meta-analysis of depression, anxiety, and suicidal ideation among Ph.D. students

[Emily N. Satinsky](#) , [Tomoki Kimura](#), [Mathew V. Kiang](#), [Rediet Abebe](#), [Scott Cunningham](#), [Hedwig Lee](#), [Xiaofei Lin](#), [Cindy H. Liu](#), [Igor Rudan](#), [Srijan Sen](#), [Mark Tomlinson](#), [Miranda Yaver](#) & [Alexander C. Tsai](#) 

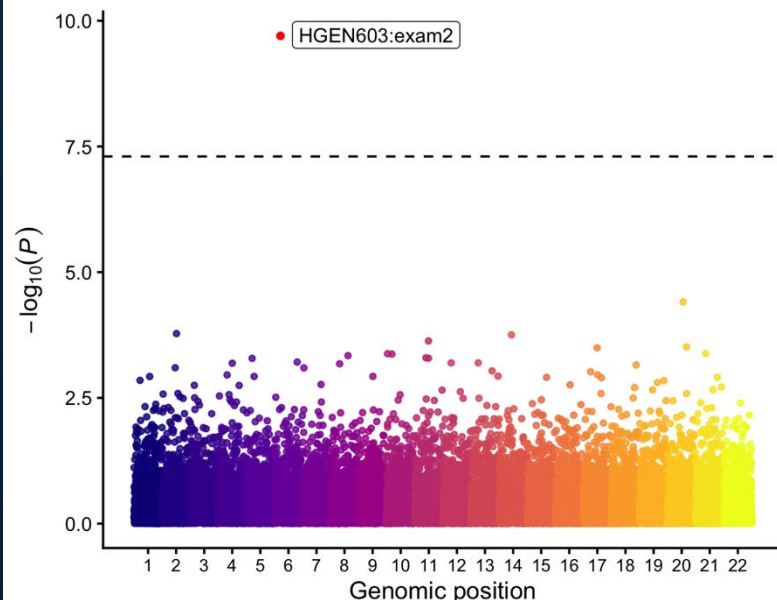
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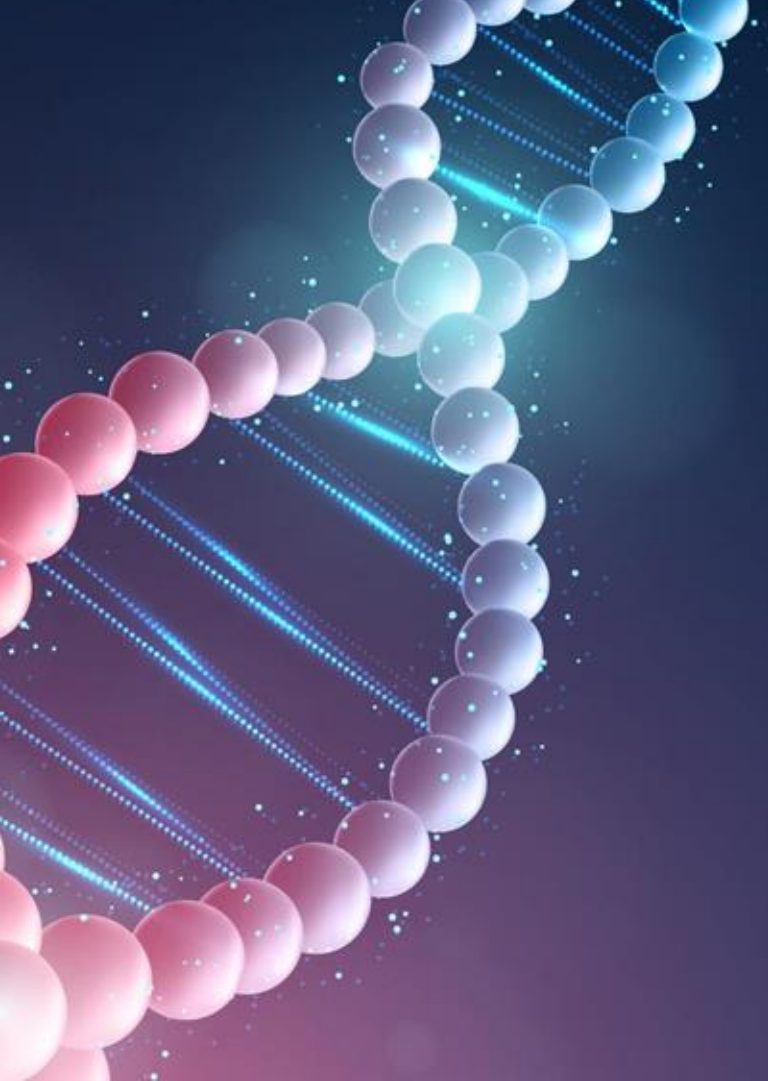
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Abstract

University administrators and mental health clinicians have raised concerns about depression and anxiety among Ph.D. students, yet no study has systematically synthesized the available evidence in this area. After searching the literature for studies reporting on depression, anxiety, and/or suicidal ideation among Ph.D. students, we included 32 articles. Among 16 studies reporting the prevalence of clinically significant symptoms of depression across 23,469 Ph.D. students, the pooled estimate of the proportion of students with depression was 0.24 (95% confidence interval [CI], 0.18–0.31; $I^2=98.75\%$). In a meta-analysis of the nine studies reporting the prevalence of clinically significant symptoms of anxiety across 15,626 students, the estimated proportion of students with anxiety was 0.17 (95% CI, 0.12–0.23; $I^2=98.05\%$). We conclude that depression and anxiety are highly prevalent among Ph.D. students. Data limitations precluded our ability to obtain a pooled estimate of suicidal ideation prevalence. Programs that systematically monitor and promote the mental health of Ph.D. students are urgently needed.

Manhattan Plot





Research Question



RQ1: Demographic Differences

Do individuals with antidepressant treatment escalation differ from those maintained on a single SSRI in **demographic characteristics** (age, sex, race, and ethnicity)?

RQ2: Clinical Complexity

Do individuals with treatment escalation differ in **psychiatric, medical comorbidities** and **family history**?

RQ3: Healthcare Utilization and Medication burden

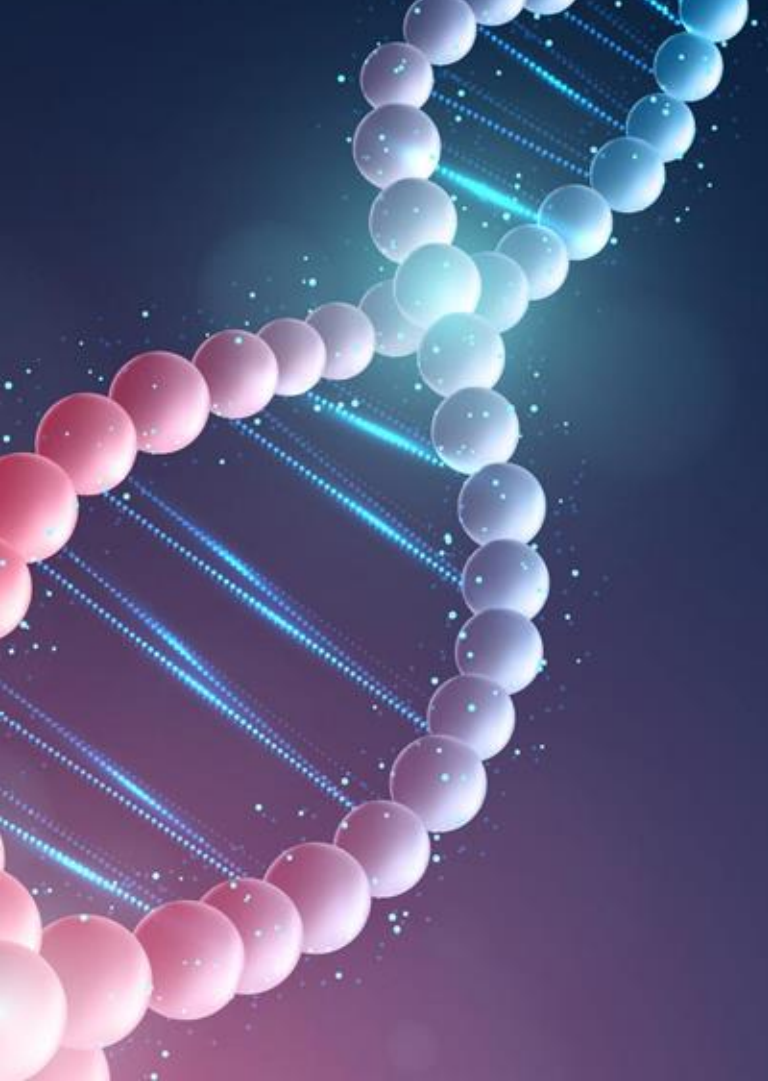
Do individuals with treatment escalation have **higher medication complexity**?

RQ4: Clinical Measurements & Severity

Do **clinical measurements** (BMI, BP, CRP, PHQ-9) differ between groups?

RQ7: Predictive Modeling

Which variables independently **predict treatment escalation (TRD)** in a multivariable model?



Methods



Data Source and Study Population: All of Us Research Program (Controlled Tier)

Phenotype Definition :

1. TRD group (treatment escalation):

a) ≥ 2 switches between antidepressant medications

OR

b) Evidence of augmentation therapy (e.g., antipsychotics, lithium, other class of antidepressant, MAOI, etc.)

OR

c) Using ketamine/esketamine alone or beside other medication
(ketamine shouldn't be used for surgery, should be used ≥ 3)

2. Non-TRD group (single treatment): Individuals with major depressive disorder, identified using ICD-9 and ICD-10 codes (e.g., F32, F33), who remained on a single antidepressant (e.g., an SSRI) without switching medications more than twice.

TRD status was coded as a binary outcome (TRD vs Non-TRD)



Covariates:

- **Demographic variables:** Age at index, Sex at birth ,Race, Ethnicity
- **Clinical comorbidities (EHR-derived):** Anxiety disorder, Bipolar disorder, Opioid use disorder (OUD), Hypertension (HTN)
- **Family history (self-reported survey data):** Depression, Anxiety, Bipolar disorder, PTSD, Substance use
- **Healthcare utilization:** Number of medical visits within one year
- **Number of unique medications**
- **Clinical measurements:** BMI, blood pressure, CRP, TSH, vitamin D, PHQ-9 score

Statistical Analysis:

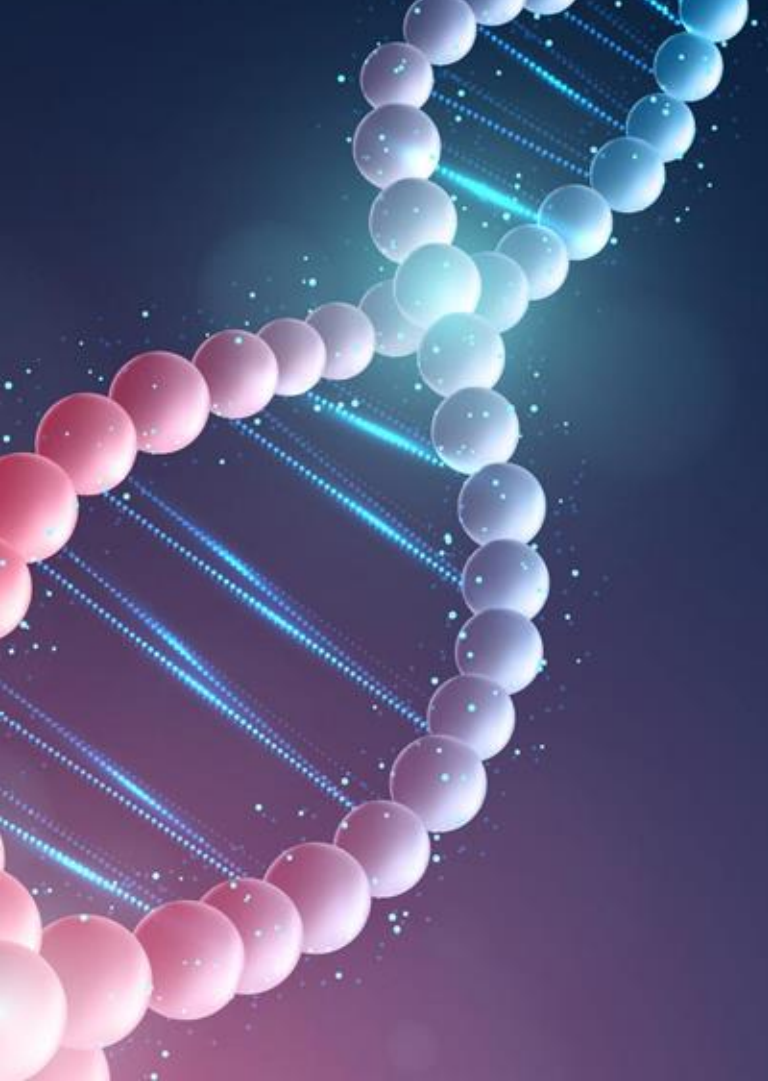
Baseline characteristics were summarized separately for TRD and Non-TRD groups using:

- a) **Counts and percentages** for **categorical** variables
- b) **Means or medians** for **continuous** variables

Group differences were assessed using:

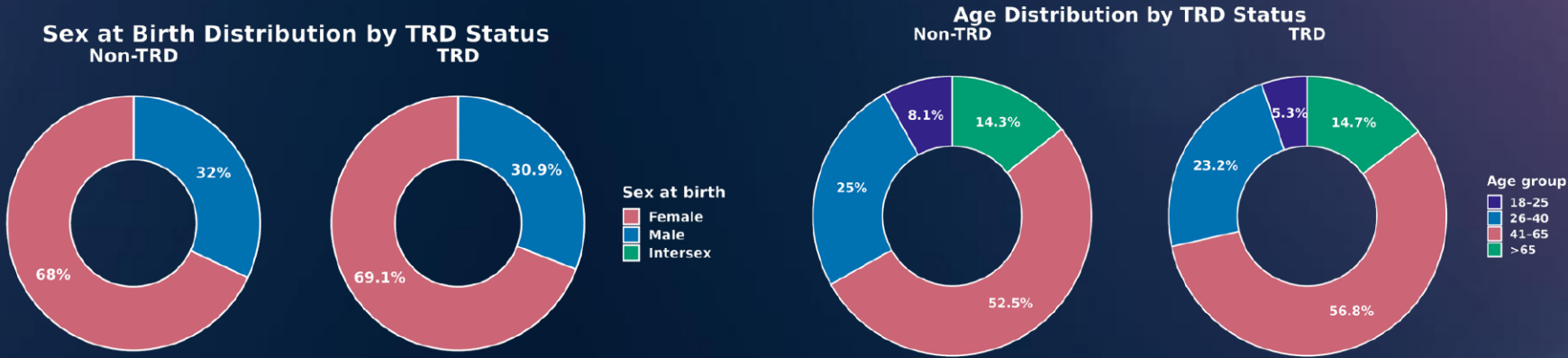
- a) **Chi-squared tests** for **categorical** variables
- b) **Wilcoxon rank-sum tests** for **continuous** variables

Logistic Regression Analysis: Model discrimination was assessed using the **area under the ROC curve (AUC)** and Multicollinearity was evaluated using **variance inflation factors (VIF)**



Results

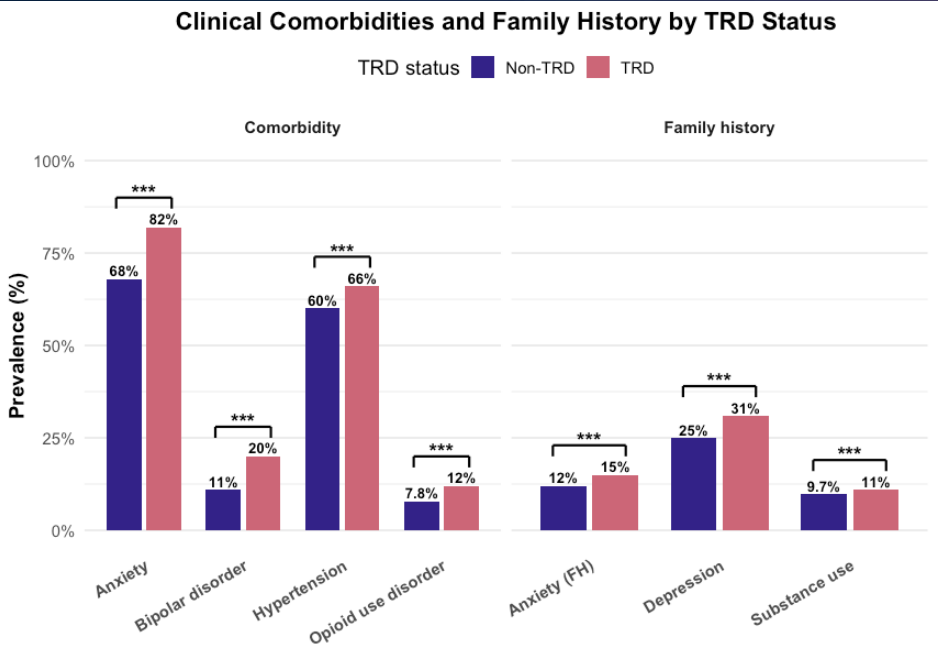
Demographic Table: Age and Sex at birth



Demographic Table: Comorbidities and Family History

Characteristic	Non-TRD N = 33,891 ¹	TRD N = 44,360 ¹	p-value ²
Anxiety			<0.001
No	10,541 (32%)	7,784 (18%)	
Yes	22,758 (68%)	36,396 (82%)	
Bipolar disorder			<0.001
No	29,741 (89%)	35,546 (80%)	
Yes	3,558 (11%)	8,634 (20%)	
Opioid use disorder			<0.001
No	30,687 (92%)	38,900 (88%)	
Yes	2,612 (7.8%)	5,280 (12%)	
Hypertension			<0.001
No	13,440 (40%)	14,996 (34%)	
Yes	19,859 (60%)	29,184 (66%)	
Family history: Depression			<0.001
No	25,342 (75%)	30,598 (69%)	
Yes	8,549 (25%)	13,762 (31%)	
Family history: Anxiety			<0.001
No	29,973 (88%)	37,740 (85%)	
Yes	3,918 (12%)	6,620 (15%)	
Family history: Substance use			<0.001
No	30,592 (90%)	39,311 (89%)	
Yes	3,299 (9.7%)	5,049 (11%)	

¹ n (%); Mean (SD)
² Pearson's Chi-squared test; Wilcoxon rank sum test



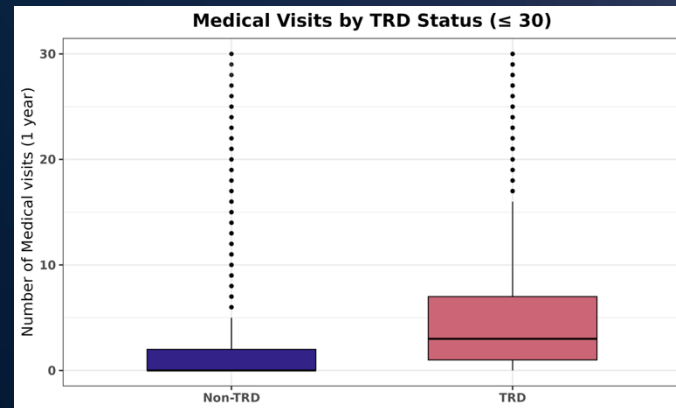
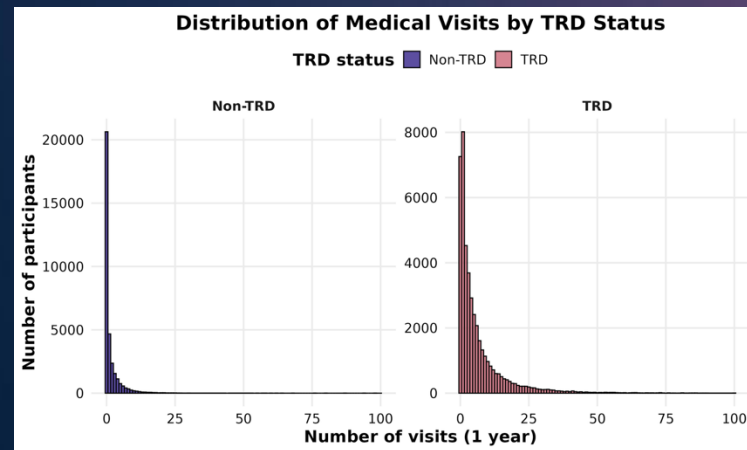
Demographic Table: Medical visits within a year

Characteristic	Non-TRD N = 33,891 ¹	TRD N = 44,360 ¹	p-value ²
Medical visits in 1 year	2 (4)	7 (11)	<0.001
Number of unique medications	1.10 (0.60)	3.67 (1.83)	<0.001

¹ n (%); Mean (SD)

² Pearson's Chi-squared test; Wilcoxon rank sum test

Metric	Non-TRD (N = 33,891)	TRD (N = 44,360)
Median (Q1–Q3)	0 (0–2)	3 (1–8)
IQR	2	7
Upper outlier cutoff (Q3 + 1.5×IQR)	5.0	18.5
Maximum visits	140	219
High outliers (n)	2,771	3,964
High outliers (%)	8.18%	8.94%
>30 visits (n, %)	104 (0.31%)	1,623 (3.66%)
>50 visits (n, %)	27 (0.08%)	501 (1.13%)
>100 visits (n, %)	4 (0.01%)	56 (0.13%)

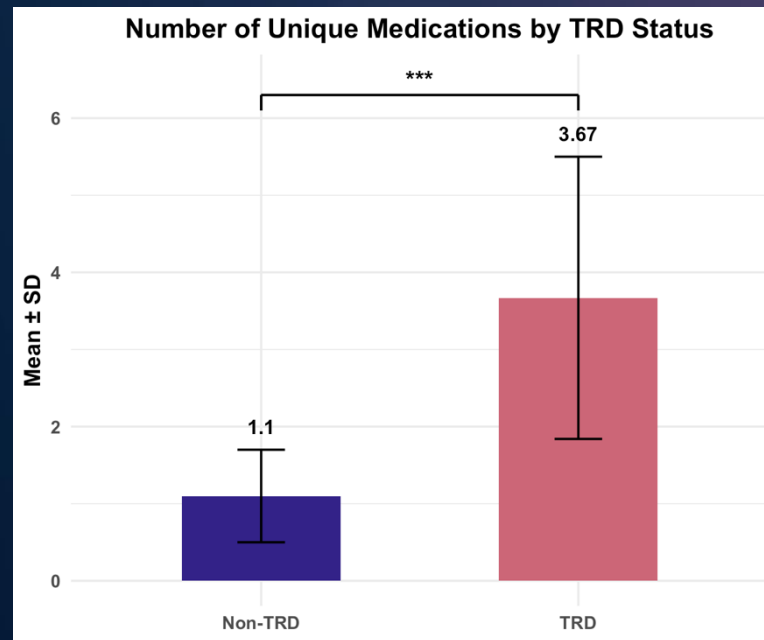


Demographic Table: Unique Medications

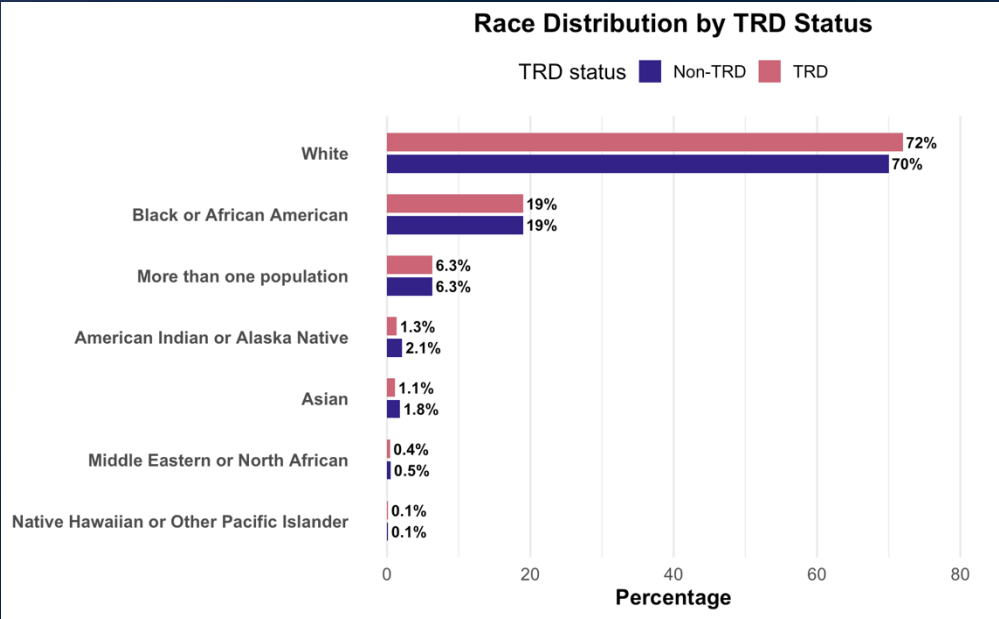
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¹ n (%); Mean (SD)

² Pearson's Chi-squared test; Wilcoxon rank sum test



Demographic Table: Race and Ethnicity



Characteristic	Non-TRD N = 33,891 ¹	TRD N = 44,360 ¹	p-value ²
Race			
<0.001			
American Indian or Alaska Native	593 (2.1%)	488 (1.3%)	
Asian	495 (1.8%)	423 (1.1%)	
Black or African American	5,416 (19%)	7,413 (19%)	
Middle Eastern or North African	145 (0.5%)	153 (0.4%)	
More than one population	1,775 (6.3%)	2,422 (6.3%)	
Native Hawaiian or Other Pacific Islander	38 (0.1%)	27 (<0.1%)	
White	19,635 (70%)	27,466 (72%)	
Ethnicity			
<0.001			
Hispanic or Latino	5,979 (18%)	5,608 (13%)	
Not Hispanic or Latino	26,919 (82%)	37,333 (87%)	

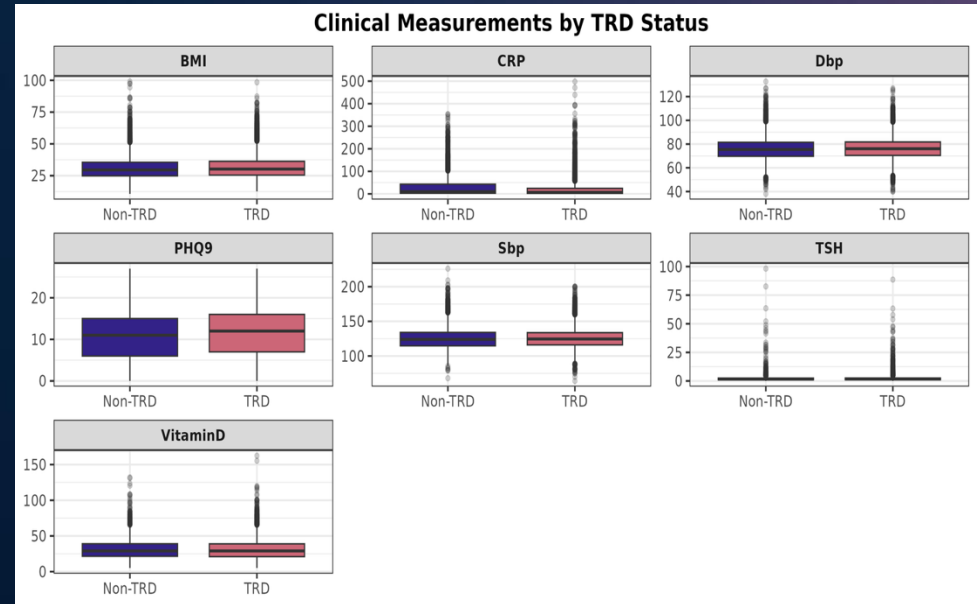
¹ n (%); Mean (SD)

² Pearson's Chi-squared test; Wilcoxon rank sum test

Summary of Measurements

Clinical Measurements by TRD Status			
Clinical measurement	Non-TRD ¹	TRD ¹	p-value ²
BMI	30 (25, 36)	30 (25, 36)	<0.001
Systolic blood pressure	124 (115, 134)	125 (116, 134)	<0.001
Diastolic blood pressure	75 (70, 82)	76 (70, 82)	<0.001
CRP	10 (3, 43)	7 (2, 25)	<0.001
TSH	1.60 (1.06, 2.45)	1.66 (1.06, 2.52)	0.010
Vitamin D	29 (22, 39)	29 (21, 39)	0.8
PHQ-9 score	11 (6, 15)	12 (7, 16)	0.001

¹ Median (Q1, Q3)
² Wilcoxon rank sum test

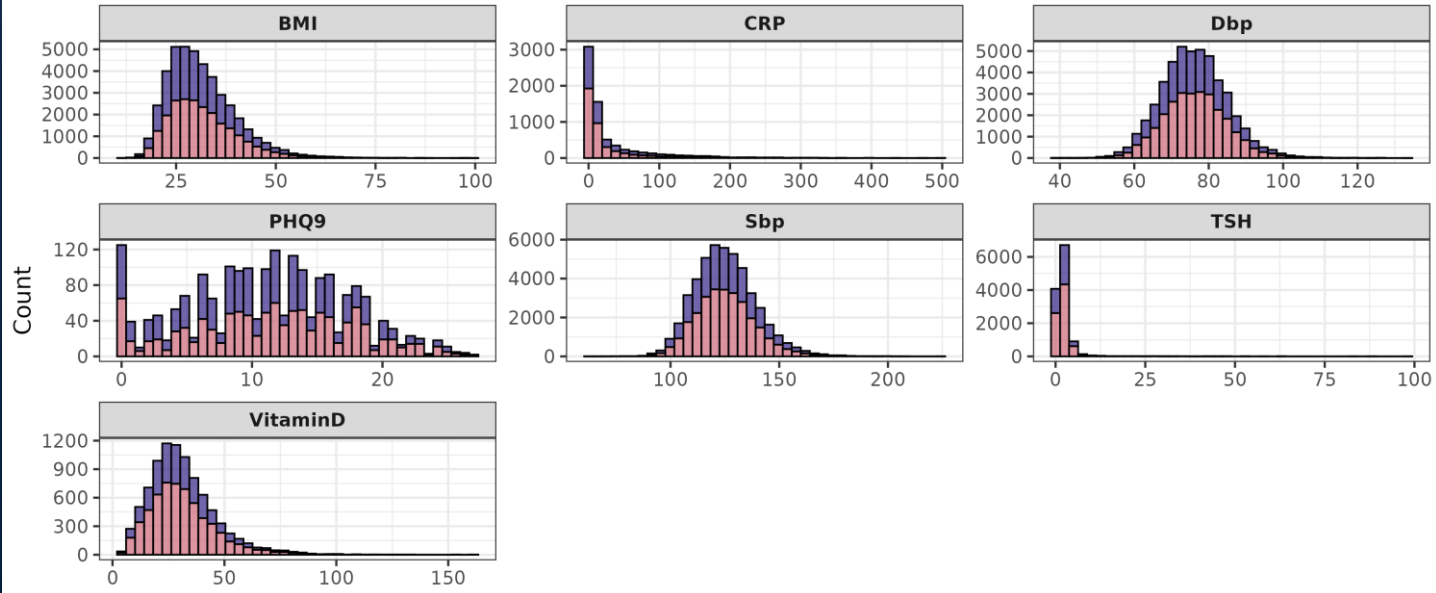


Although several variables were statistically different, most differences were small and may not be clinically meaningful.

Summary of Measurements

Distribution of Clinical Measurements

TRD status ■ Non-TRD ■ TRD

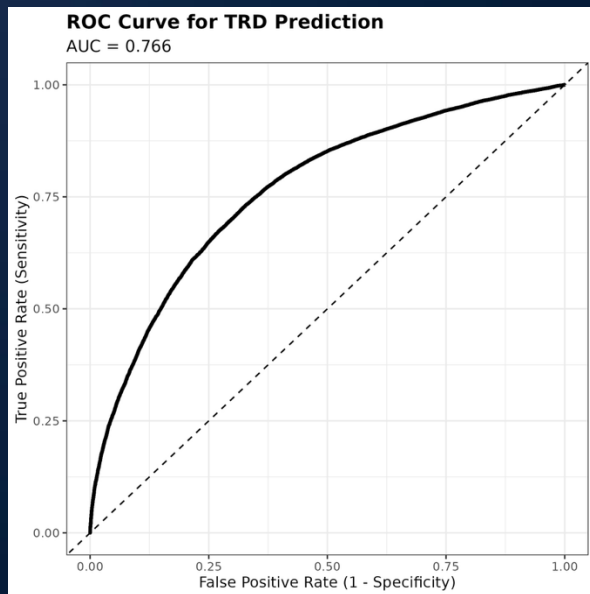


Association test: Logistic Regression Model

No Multicollinearity

$TRD \sim age + sex_at_birth + race + ethnicity + has_anxiety + has_bipolar + has_oud + has_htn + fh_depression + fh_anxiety + fh_bipolar + fh_ptsd + fh_substance + log_visits$

Characteristic	OR	95% CI	p-value
Log(visits)	2.60	2.55–2.66	<0.001
Bipolar disorder (Yes vs No)	1.94	1.84–2.05	<0.001
Anxiety (Yes vs No)	1.80	1.73–1.88	<0.001
Opioid use disorder (Yes vs No)	1.46	1.37–1.56	<0.001
More than one population (race)	1.42	1.21–1.66	<0.001
White (race)	1.41	1.22–1.63	<0.001
Black or African American (race)	1.41	1.21–1.64	<0.001
Ethnicity: Not Hispanic	1.28	1.15–1.42	<0.001
Family history: Depression	1.23	1.18–1.30	<0.001
Family history: PTSD	1.23	1.12–1.35	<0.001
Hypertension (Yes vs No)	1.14	1.09–1.18	<0.001
Family history: Anxiety	1.13	1.06–1.22	<0.001
Middle Eastern/North African	1.09	0.81–1.48	0.56
Native Hawaiian / Pacific Islander	1.03	0.55–1.90	0.93
Age (per year)	1.01	1.01–1.01	<0.001
Family history: Bipolar	0.98	0.89–1.07	0.64
Asian	0.95	0.77–1.18	0.67
Family history: Substance	0.95	0.88–1.01	0.12
Intersex vs Female	0.94	0.35–2.65	0.91
Male vs Female	0.81	0.78–0.85	<0.001



Systematic review and meta-analysis of depression, anxiety, and suicidal ideation among Ph.D. students

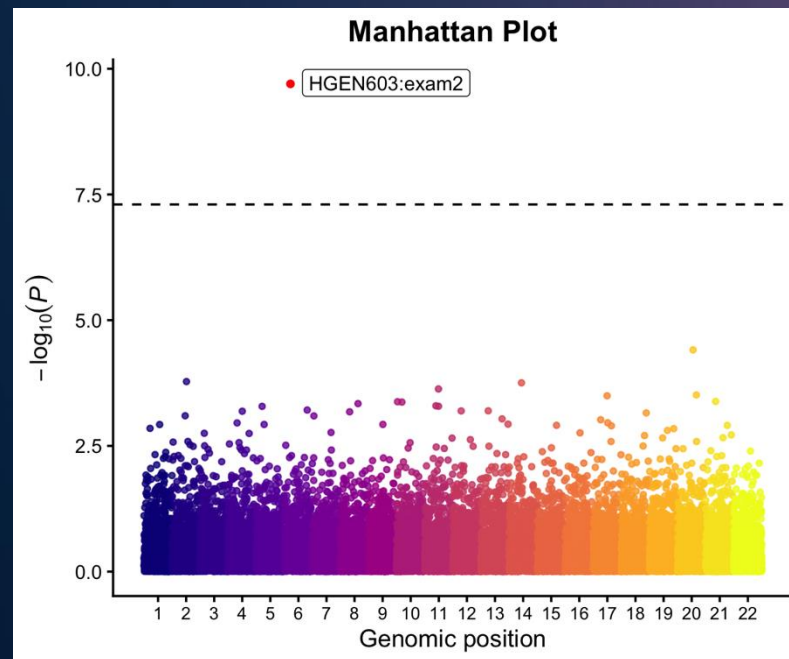
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Abstract

University administrators and mental health clinicians have raised concerns about depression and anxiety among Ph.D. students, yet no study has systematically synthesized the available evidence in this area. After searching the literature for studies reporting on depression, anxiety, and/or suicidal ideation among Ph.D. students, we included 32 articles. Among 16 studies reporting the prevalence of clinically significant symptoms of depression across 23,469 Ph.D. students, the pooled estimate of the proportion of students with depression was 0.24 (95% confidence interval [CI], 0.18–0.31; $I^2=98.75\%$). In a meta-analysis of the nine studies reporting the prevalence of clinically significant symptoms of anxiety across 15,626 students, the estimated proportion of students with anxiety was 0.17 (95% CI, 0.12–0.23; $I^2=98.05\%$). We conclude that depression and anxiety are highly prevalent among Ph.D. students. Data limitations precluded our ability to obtain a pooled estimate of suicidal ideation prevalence. Programs that systematically monitor and promote the mental health of Ph.D. students are urgently needed.





Discussion



The gist:

- Among about 78,000 individuals with major depressive disorder, **56.6% met criteria for treatment escalation**.
- **Demographic differences between groups were modest**. The TRD group was slightly older and had a higher proportion of non-Hispanic individuals. Male sex was associated with lower odds of TRD
- **Clinical characteristics differed substantially**, with **higher prevalence of psychiatric comorbidities** in the TRD group
- **Family history of depression and anxiety** was also more common among TRD individuals.
- **Healthcare utilization showed the largest differences**, with TRD patients having significantly higher visit frequency (median 3 vs 0 visits/year) and a greater proportion of high-utilization outliers.
- TRD patients also had a **higher number of unique medications** (mean 3.67 vs 1.10).
- Although several clinical measurements (BMI, blood pressure, CRP, PHQ-9) were statistically different, the magnitude of these differences was small and may not be clinically meaningful.
- In multivariable logistic regression, **healthcare utilization was the strongest predictor of TRD** followed by bipolar disorder (OR = 1.94) and anxiety (OR = 1.80). The model demonstrated good discrimination (AUC = 0.766).



Future direction:

Do more data cleaning, then run a GWAS or PRS analysis

Some Challenges Encountered During the Project:

1. There is **no specific ICD code for treatment-resistant depression (TRD)**, so it must be defined indirectly using EHR data.
2. **Cleaning medication** data and accurately calculating the number of **psychiatric drug switches**
3. **Age** calculation
4. The **dataset is very large**, and loading or processing it sometimes caused crashes or performance issues

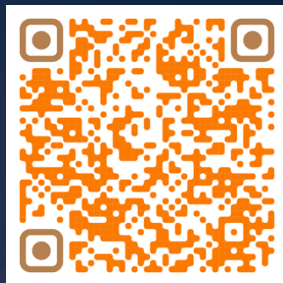
Advice:

1. For each step, create a separate dataset and save it as an **.rds file**. This way, you don't need to rerun all the code again.
2. Define your research question and phenotype clearly and specifically at the beginning. The more **precise your definition**, the easier it will be to **clean and analyze** the data.
3. Take **annotations** and **headings** seriously. They may seem time-consuming or unnecessary at first, but they become extremely helpful later when revisiting or debugging your code.
4. At the final step of cleaning each dataset, save a clean, finalized version and **remove unnecessary columns** to keep your workflow efficient and organized.

References

1. Zhdanava M, Pilon D., et al. The Prevalence and National Burden of Treatment-Resistant Depression and Major Depressive Disorder in the United States. *J Clin Psychiatry*. 2021 Mar 16;82(2):20m13699. PMID: 33989464.
2. McIntyre RS, Alsuwaidan M., et al. Treatment-resistant depression: definition, prevalence, detection, management, and investigational interventions. *World Psychiatry*. 2023 Oct;22(3):394-412. doi: 10.1002/wps.21120. PMID: 37713549; PMCID: PMC10503923.
3. Kajumba, Mayanja M., et al. "Treatment-resistant depression: molecular mechanisms and management." *Molecular Biomedicine* 5.1 (2024): 1-29.
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5. Kim, Yong-Ku. "Recent Advances and Challenges in the Treatment of Major Depressive Disorder." (2024).
6. <https://www.uptodate.com/contents/unipolar-depression-in-adults-choosing-treatment-for-resistant-depression>

**If you want to check whether you
are depressed enough or not**



HAM-D



PHQ-9



DSM-5

Thank you for listening

"life is full of ups and downs, but you live once, so it is worth fighting for"